

Aryamaan Dash

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EDUCATION

University of California, Irvine

Bachelor of Science in Computer Science & Engineering, Minor in Mathematics

Irvine, CA

June 2028

EXPERIENCE

Software Engineer Intern

March 2024 – December 2024

Hey, Blue!

Remote

- Built and shipped 6 React pages and UI components with React.js, HTML, and CSS across desktop and mobile layouts.
- Led a 5-developer front-end team, breaking product requirements into scoped, trackable implementation tasks.
- Coordinated feature ownership and tracked deliverables from initial development through review and release.

PROJECTS

Study Tracker | *Next.js, TypeScript, Auth.js, Prisma, PostgreSQL, Docker*

Apr. 2026 – Sep. 2026

- Built a full-stack study tracker with Next.js, TypeScript, PostgreSQL, Prisma, and secure Auth.js authentication.
- Developed interactive charts, calendars, focus timers, journal entries, and weekly reflections to visualize learning progress.
- Optimized performance with parallel data fetching, indexed queries, and tag-based caching; added automated testing and Dockerized development.

Programmable Multi-Effects Pedal | *C++, STM32, DaisySP, KiCad*

Jun. 2026 – Sep. 2026

- Built a Daisy Seed-based programmable guitar pedal with selectable bypass, distortion, reverb, delay, and flanger effects.
- Programmed real-time embedded C++ DSP firmware using DaisySP, audio callbacks, delay-line processing, wet/dry mixing, and debounced mode switching.
- Tested the pedal with 15 users, receiving positive ratings for its sound quality, effects, and uniqueness.

Reusable Embedded C Library | *C, STM32, STM32CubeMX, Cmake*

Aug. 2026

- Developed a reusable embedded C library with 5 core modules: GPIO, UART, ring buffer, FSM, and button debouncing using defensive input validation and static/caller-provided memory for embedded environments.
- Designed a platform abstraction layer separating hardware-independent GPIO/UART APIs from STM32G071 HAL implementations, translating portable driver types and status codes to platform-specific HAL operations.
- Implemented an O(1) circular byte buffer, configurable time-based button debouncing with press/release events, and a table-driven state machine that validates states, events, and duplicate transitions before execution.

Seam Carving Web App | *Next.js, React, Node.js, TypeScript, Python, Numpy*

Jun. 2026 – Jul. 2026

- Built and deployed a full-stack image resizing web app with upload previews, axis controls, projected dimensions, and downloadable results.
- Engineered an API route that validates PNG, JPEG, and WebP uploads up to 12 MB, creates isolated temporary jobs, invokes the Python processor, enforces timeouts, and cleans up files.
- Implemented content-aware resizing using Sobel energy maps, dynamic programming, and seam backtracking to remove low-energy pixels while preserving image content

TECHNICAL SKILLS

Languages: C/C++, Python, SQL (Postgres), TypeScript, HTML/CSS, Verilog

Frameworks: Node.js, Flask, Pytorch, Tensorflow, Flask

Developer Tools: Git, Docker, Google ADK, VS Code, Linux, Vivado, STM32CubeMX

Libraries: React, pandas, NumPy, Matplotlib, Keras, scikit-learn, GoogleTest